Lines Drawn in 1937

84-355 Democracy's Data | HOLC grades against the 2020 census

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Everything here runs as given. Your job is to read what comes out and say what it means.

> **One of three labs in this unit.** All three ask the same question of different data: **can you show discrimination when nobody wrote down a reason?** The others are policing and jury selection. Whichever you did, you will be asked to explain your inference to somebody who did a different one.
>
> **Read `redlining-brief.Rmd` first.** This lab runs the same numbered steps with the code exposed. The brief tells you what the question is and why it is hard; this shows you where each number came from.

Where did that number come from?

> *"You can still see the redlining maps in the census data."*

In the late 1930s the **Home Owners' Loan Corporation** — a federal agency — graded neighbourhoods in 200-odd American cities for mortgage risk. **A** was "Best", green on the map. **D** was "Hazardous", red. The written appraisals name the presence of Black residents as a reason for a low grade, in so many words.

The maps were drawn ninety years ago. The claim above says they are still visible. **That is a testable claim**, and unlike the other two labs in this unit, here the discriminatory intent is not hidden at all — it is written on the document. What has to be shown is that it *still matters*.

Steps 1-3 — Two sources, the join that has to be built, and one record

Two datasets, drawn eighty years apart, that were never meant to meet:

- **HOLC grades**, digitised by the University of Richmond: 10,154 areas drawn on the 1930s security maps, 9,324 of them carrying a grade from A to D.
- **The 2020 census**, by tract.

Neither carries the other's identifier. To connect them you have to decide **when a tract counts as being in a graded area** — and we did the simplest thing: take the tract's centre point and ask which polygon it falls inside.

```
# GEOID is text, never a number. A tract identifier is 11 digits and a third of
# the ones here start with a zero (California is state 06). Read as a number
# they come back 10 digits long and match nothing.
tr <- read.csv("data/tracts.csv", stringsAsFactors = FALSE,
               colClasses = c(GEOID = "character"))
ci <- read.csv("data/cities.csv", stringsAsFactors = FALSE)

c(tracts_matched = nrow(tr), cities = length(unique(tr$city)))
```

```
#> tracts_matched         cities
#>           4489             56
```

```
table(tr$grade)
```

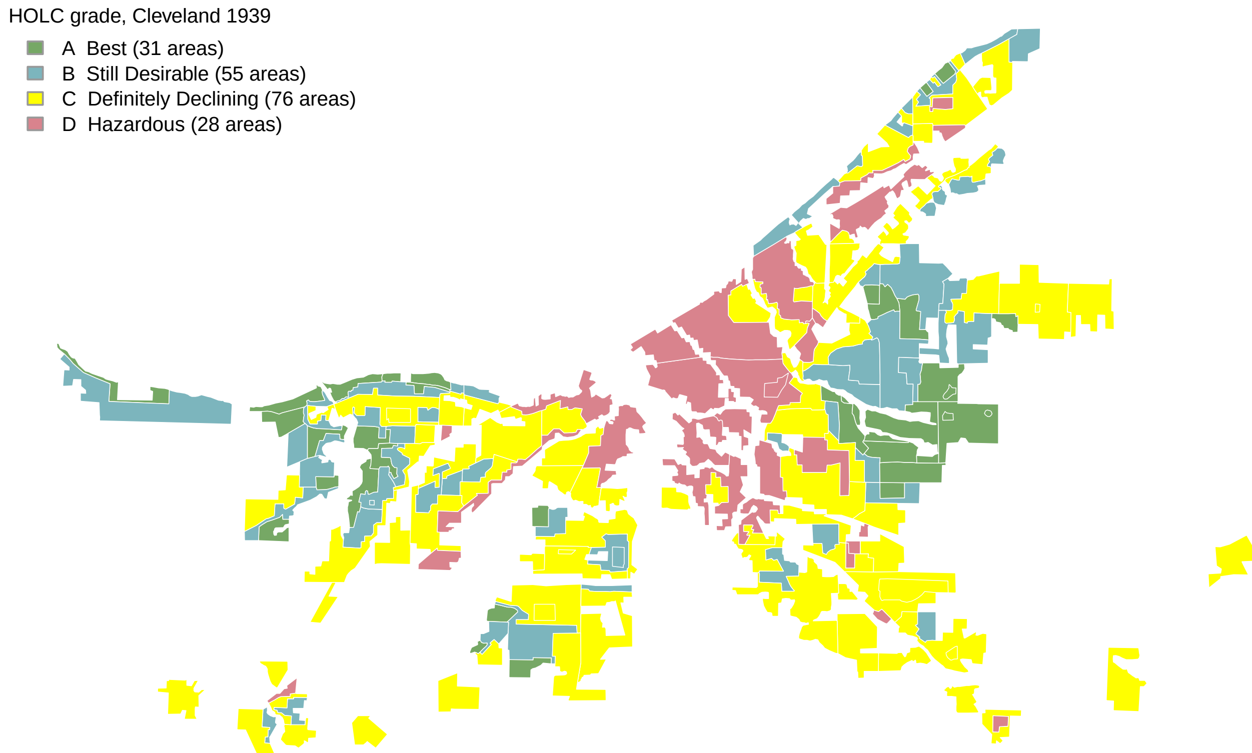
```
#>
#>    A    B    C    D
#>  273  899 2096 1221
```

The map the grades came off

Before any of that, look at the thing itself. These are the 1939 HOLC areas for Cleveland, in the four colours the HOLC printed them in — the colour of each area is read out of the source file, not chosen by us.

```
rings <- read.csv("data/fig_holc_rings.csv", stringsAsFactors = FALSE)
areas <- read.csv("data/fig_holc_attr.csv", stringsAsFactors = FALSE)
cols  <- setNames(areas$fill[match(c("A", "B", "C", "D"), areas$grade)],
                  c("A", "B", "C", "D"))

par(mar = c(0.1, 0.1, 0.1, 0.1))
plot(NA, xlim = range(rings$x), ylim = range(rings$y), asp = 1,
     axes = FALSE, ann = FALSE)
for (i in areas$id) {
  d <- rings[rings$id == i, ]
  for (q in unique(d$part))
    polygon(d$x[d$part == q], d$y[d$part == q],
            col = cols[[areas$grade[areas$id == i]]], border = "white", lwd = 0.4)
}
legend("topleft", sprintf("%s  %s (%d areas)", names(cols),
       c("Best", "Still Desirable", "Definitely Declining", "Hazardous"),
       as.vector(table(areas$grade)[names(cols)])),
       fill = cols, border = "#999999", bty = "n", cex = 0.7,
       title = "HOLC grade, Cleveland 1939", title.adj = 0)
```



Red in the middle, green at the edge. Every number in this lab is a comparison between the red parts of maps like this one and the green parts, eighty years later. The polygons come from Mapping Inequality; the brief draws the same map with the 2020 tract boundaries laid over it, which is where you can see how badly the two geographies fit.

The crosswalk matches **11,274 tracts across 215 cities** to a HOLC grade nationally. **4,489 of them, in 56 cities, also have census data here**, because we pulled the 2020 tract file for six states — PA, IL, MI, OH, MO and CA.

Every one of those matches is a judgement call: a tract whose centre falls just outside a graded area gets nothing, and a tract straddling an A and a D gets whichever polygon contains its centre.

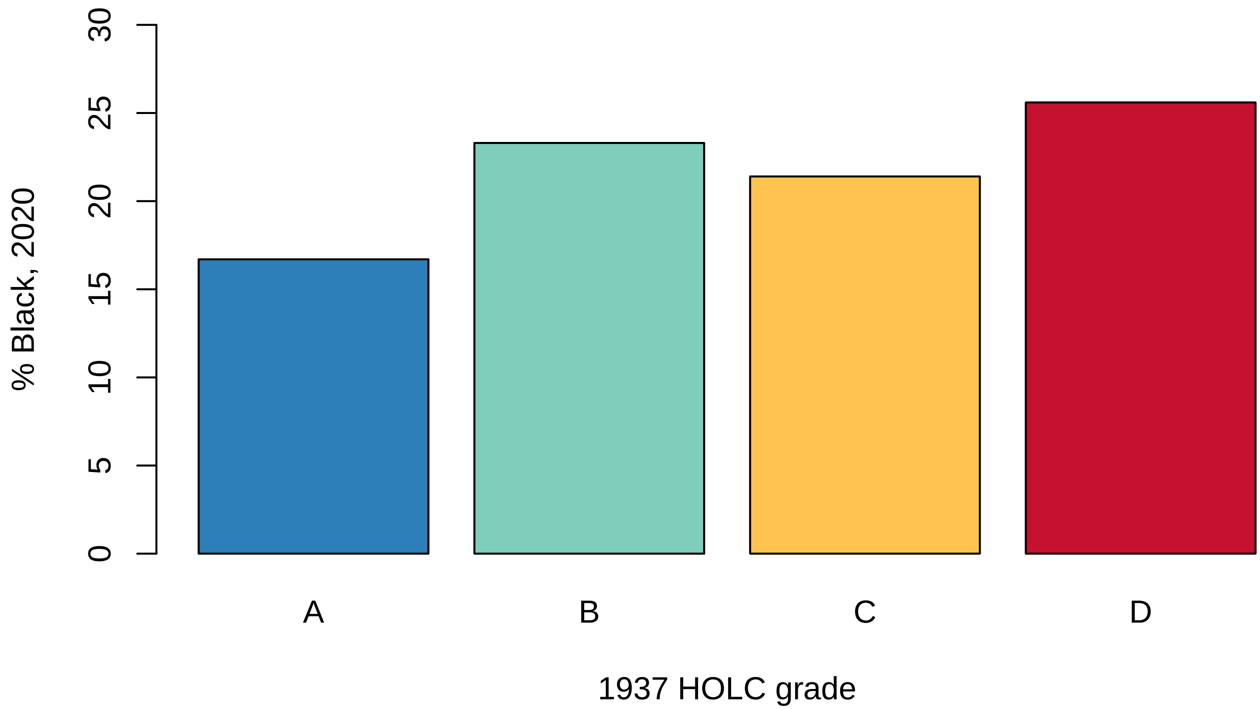
Hold on to that. It is the same allocation problem as precincts and census blocks, and there is no correct answer here either.

Steps 4-5 — The first comparison, and what is wrong with it

```r
g <- aggregate(cbind(total, black) ~ grade, data = tr, sum)
g$pct_black <- round(100 * g$black / g$total, 1)
g
```

```
#>   grade   total   black pct_black
#> 1     A  978270  163372      16.7
#> 2     B 3269467  762053      23.3
#> 3     C 7273922 1555044      21.4
#> 4     D 3984323 1018080      25.6
```

```r
barplot(g$pct_black, names.arg = g$grade, col = c("#2c7fb8","#7fcdbb","#fec44f","#C41230"),
        ylab = "% Black, 2020", xlab = "1937 HOLC grade", ylim = c(0, 30))
```



Tracts graded **D** in 1937 are **25.6%** Black today. Tracts graded **A** are **16.7%**.

That is a real gap and a disappointing one — a ratio of about 1.5. If you expected the redlining maps to leap out of the census, this looks like weak evidence. Plenty of published claims stop here, in one direction or the other.

They should not. The number is wrong, and it is wrong for a reason this course has met before.

Steps 6-7 — Compare within cities, and every city at once

> **Before running this, work out what was wrong with the last comparison.** It put A-graded areas in one city beside D-graded areas in another. Cities differ enormously in overall Black population for reasons that have nothing to do with the HOLC. **What is the comparison that holds the city constant?**

That table adds up cities. Chicago and Detroit and Los Angeles have very different racial compositions **overall**, for reasons that have nothing to do with which tracts got which grade. Pooling them compares a D-graded tract in one city against an A-graded tract in another.

The comparison has to happen inside a city.

```r
head(ci[, c("city", "a_pct_black", "d_pct_black", "gap")], 8)
```

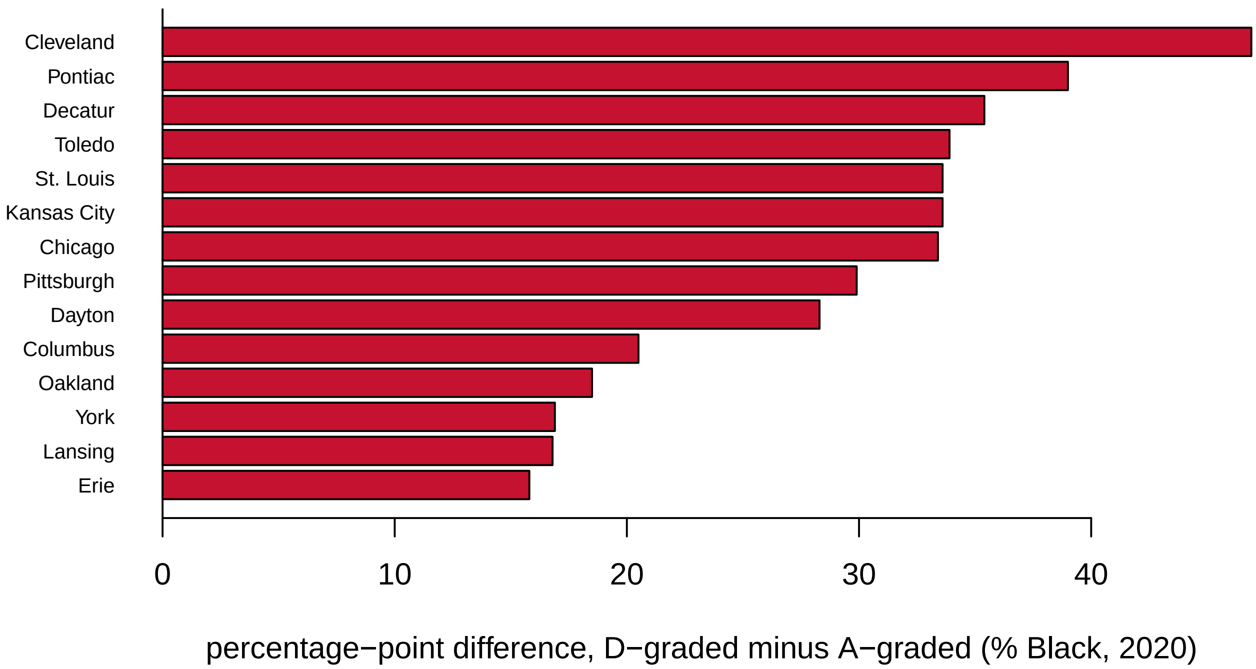
```
#>                  city a_pct_black d_pct_black  gap
#> 1           Cleveland        21.3        68.1 46.9
#> 2             Pontiac        32.1        71.0 39.0
#> 3             Decatur        19.4        54.8 35.4
```

```
#> 4             Toledo         6.6        40.5 33.9
#> 5 Greater Kansas City         4.0        37.7 33.6
#> 6           St. Louis        14.0        47.6 33.6
#> 7             Chicago         3.2        36.6 33.4
#> 8          Pittsburgh         4.2        34.1 29.9
```

```
tail(ci[, c("city", "a_pct_black", "d_pct_black", "gap")], 5)
```

```
#>            city a_pct_black d_pct_black  gap
#> 29      Detroit        46.8        43.9 -3.0
#> 30       Canton        15.2        11.7 -3.5
#> 31     Stockton        11.5         6.3 -5.2
#> 32 Philadelphia        40.7        34.3 -6.4
#> 33       Joliet        16.0         6.3 -9.6
```

```
k <- head(ci[order(-abs(ci$gap)), ], 14)
k <- k[order(k$gap), ]
barplot(k$gap, horiz = TRUE, names.arg = k$city, las = 1, cex.names = 0.7,
        col = ifelse(k$gap > 0, "#C41230", "#2c7fb8"),
        xlab = "percentage-point difference, D-graded minus A-graded (% Black, 2020)")
abline(v = 0)
```



Now it is visible, and it is enormous.

- **Cleveland:** A-graded tracts are 21.3% Black. D-graded, **68.1%**.
- **Pittsburgh:** 4.2% against **34.1%** — eight times.
- **Chicago:** 3.2% against **36.6%** — eleven times.

The national number said 1.5×. Within cities it reaches ten. Pooling did not weaken the finding slightly; it nearly erased it.

Steps 8-9 — Take the reversals seriously

```
ci[ci$gap < 0, c("city", "a_pct_black", "d_pct_black", "gap")]
```

```
#>             city a_pct_black d_pct_black  gap
#> 28      San Jose         5.4         3.7 -1.7
#> 29       Detroit        46.8        43.9 -3.0
#> 30        Canton        15.2        11.7 -3.5
#> 31      Stockton        11.5         6.3 -5.2
#> 32  Philadelphia        40.7        34.3 -6.4
#> 33        Joliet        16.0         6.3 -9.6
```

In some cities the relationship runs **backwards**. Philadelphia's A-graded tracts are **40.7%** Black and its D-graded tracts **34.3%**.

Do not explain this away. It is the most interesting row in the lab, and the explanation is history rather than error: in cities that lost enormous white population after 1950, the neighbourhoods graded "Best" in 1937 — leafy, desirable, close in — are exactly the ones that later became majority Black. The 1937 grade stopped predicting because the city around it changed completely.

A finding that holds in most places and reverses in some is not a broken finding. It is a finding with a mechanism, and the reversals are where you can see the mechanism working.

Step 10 — What you can now say

You can say that within cities, neighbourhoods the federal government graded "Hazardous" in the 1930s are on average **14.5 points more Black today** than neighbourhoods it graded "Best" — visible eighty years and several complete rebuildings of American cities later.

You can say the pooled national comparison is misleading, and that the correct unit of comparison is the city.

You cannot say the grades caused it. You cannot say the relationship is uniform — six cities reverse, and Philadelphia reverses because North Philadelphia gentrified. And you cannot generalise beyond the cities in this file.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Weight by population. The city table treats each city's A-graded and D-graded tracts as blocks. Recompute the national figure as the *average of the within-city gaps* instead of the pooled ratio from Step 4. How different is it, and which number should a newspaper print?

Option B — B and C. Parts 3 and 4 use only A and D — the extremes. Bring in the middle grades and see whether the gradient is smooth. If B and C sit neatly between, that is evidence the grade itself mattered; if they do not, say what else might be going on.

Option C — Attack the join. Every result here depends on centroid-in-polygon. Name two ways that choice could produce the Step 6 result even if HOLC grades had no lasting effect at all. Then say what data you would need to rule them out.

Option D — Your own question. 11,274 tracts, 215 cities, one 1937 map.

What this lab cannot tell you

- **That redlining caused any of this.** The maps recorded segregation that already existed as well as entrenching it. Separating the two needs a research design this lab does not have — and it is the central difficulty of the whole literature.
- **Anything about people.** Tract composition is not any individual's experience, and inferring one from the other is the ecological fallacy the RPV lab is about.
- **The full country.** Census data here covers six states, chosen for their HOLC cities. The crosswalk covers 215 cities; the join covers fewer.
- **What "A" meant.** The grades bundled housing age, construction quality, income and race together. The written appraisals separate them; the letter does not.

On the discussion board

By **Friday, 11:59 p.m.** A sentence or two is enough.

Three that would make good posts:

- The pooled number said 1.5×; within Chicago it is eleven. What does that do to your trust in national averages?
- In Philadelphia the pattern reverses. Is that evidence against the redlining story, or evidence for a more specific version of it?
- If you did one of the other two labs in this unit, which inference did you find more convincing, and why?

Where the data came from

Mapping Inequality, Digital Scholarship Lab, University of Richmond — the digitised HOLC residential security maps, 10,154 areas of which 9,324 carry a grade, published CC BY-NC-SA. https://dsl.richmond.edu/panorama/redlining/ The polygons drawn above, the four grade colours, and the appraisal text quoted in the brief are all theirs, retrieved **2026-08-10** from `static/mappinginequality.json`. The maps and the appraisal forms underneath are 1930s US federal government records. `data/build-brief-figures.R` is the script that fetched them and measured them against the tract boundaries.

2020 Census PL 94-171, tract level, for six states (PA, IL, MI, OH, MO, CA).

The crosswalk between them is ours. A published tract-level redlining score exists — Meier & Mitchell, openICPSR 141121 — but it is behind a login. So `data/build-data.py` builds the join directly: tract centre points from the Census Gazetteer, tested against each HOLC polygon by ray casting, in about fifteen lines. You can read exactly what the join assumes, which is the point.